O-P-566B
October 30, 1984
SUPERSEDING
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FEDERAL SPECIFICATION

POTASSIUM HYDROXIDE, TECHNICAL

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers technical grade potassium hydroxide (KOH).

1.2 Classification. Potassium hydroxide shall be of the following classes, as specified (see 6.2):

Class 1 - Flakes
Class 2 - Lumps
Class 3 - Pellets
Class 4 - Sticks

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals
PPP-C-2020	-	Chemicals, Liquid, Dry, and Paste: Packaging Of
PPP-D-729	-	Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

FSC 6810

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Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Washington, DC; Philadelphia, PA; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Code of Federal Regulations (CFR):

49 CFR 171 to 179 - Hazardous Materials Regulations

(The Code of Federal Regulations and the Federal Register (FR) are for sale on a subscription basis by the Superintendant of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM Standards:

D 1193 - Reagent Water
E 291 - Chemical Analysis of Caustic Soda and Caustic Potash (Sodium Hydroxide and Potassium Hydroxide)
E 29 - Indicating Which Places of Figures are to be Considered Significant in Specified Limiting Values

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except the Code of Federal Regulations), the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Manufacture. Potassium hydroxide shall be manufactured by the electrolytic method. The contractor shall certify that this requirement has been met.

3.2 Color, form and contamination. Potassium hydroxide shall be visibly free from foreign matter, shall have a white or pale yellow color, and shall have a form conforming to table I when tested as specified in 4.2.4.1.

3.3 Chemical characteristics. Potassium hydroxide shall conform to the chemical characteristics of table II when tested as specified therein.

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TABLE I. Form

Class	Form	Dimensions
1	Flakes	----
2	Lumps	Larger than 1/2 inch and smaller than 2 inches
3	Pellets	1/4-inch maximum for any dimension
4	Sticks	1/2-inch maximum diameter

TABLE II. Chemical characteristics

Characteristic	Requirement	Test paragraph
Potassium hydroxide, percent by weight, minimum	85	4.2.4.2
Carbonate (as K_2CO_3), percent by weight, maximum	3	4.2.4.2

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the potassium hydroxide of one class, produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-2.

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4.2.2.2 For potassium hydroxide test. See 6.4 for sampling and testing precautions. Sampling shall be conducted in accordance with table III. A representative specimen of approximately 50 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE III. Sampling for potassium hydroxide test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-2.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Fiberboard liners, partitions, or pads missing or not as specified (when required)
(f) Marking incorrect, missing or illegible
(g) Unitization not as specified

4.2.3.2 For potassium hydroxide test. See 6.4 for sampling and testing precautions. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.5 percent defective.

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4.2.3.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

4.2.4 Potassium hydroxide tests. Water in accordance with ASTM D 1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Color, form, and contamination. Visually examine the specimen for foreign matter, color, and form.

4.2.4.2 Potassium hydroxide and carbonate. Determine the percentages of potassium hydroxide and carbonate in the specimen in accordance with the applicable portions of the test for caustic potash in ASTM E 291.

4.2.5 Container leakage test. Place the container in each of the following positions, and leave it in each for a period of 15 minutes.

(a) Upright
(b) Upside down
(c) On one side (or one quadrant)
(d) On one end (or second quadrant)
(e) On other side (or fourth quadrant)

Examine the container after each period for any evidence of leakage.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or C as specified (see 6.2), and shall be in accordance with Department of Transportation (DOT) regulations and the general requirements of PPP-C-2020.

5.1.1 Level A.

5.1.1.1 Unit packing. Potassium hydroxide shall be unit packed level A in a 1-, 5-, or 100-pound (lb) quantity as specified (see 6.2).

5.1.1.1.1 One-lb quantity. A quantity of 16 (+1/8 or -0) ounces (oz) of potassium hydroxide shall be unit packed in a wide-mouth glass bottle or metal can as specified (see 6.2). The glass bottle shall conform to group A, class 1, type of glass optional, style 2 wide-mouth or style 4 jar, grade of light penetration optional, closure A or R with seal A or B of PPP-C-186. The metal can shall conform to type V round, class 2, formed of No. 100 minimum electrolytic tin plate, with exterior coating plan B with side seam striped of PPP-C-96. Each container shall be closed in accordance with its respective specification. There shall be no evidence of leakage when tested as specified in 4.2.5.

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5.1.1.1.2 Five-lb quantity. A quantity of 5 lb (+1 or -0 oz) of potassium hydroxide shall be unit packed in a metal can in the same manner as specified for the 1-lb quantity in a can in 5.1.1.1.1. There shall be no evidence of leakage when tested as specified in 4.2.5.

5.1.1.1.3 One hundred-lb quantity. A quantity of 100 (+1 or -0) lb of potassium hydroxide shall be unit packed in a bag within a metal drum with removable head. The drum shall conform to DOT Specification 37A, and the exterior of the drum shall be finished as specified for the type V drum of PPP-D-729. The bag shall conform to type I, class B, style optional of MIL-B-117, and shall be closed by tying, knotting or heat sealing. The drum shall be closed tightly in accorance with the drum manufacturer's instructions. There shall be no evidence of leakage of contents from the drum when tested as specified in 4.2.5.

5.1.1.2 Intermediate packing.

5.1.1.2.1 One-lb quantity. Twelve 1-lb unit packs of one type of bottles or cans shall be intermediately packed in three rows of four unit packs per row in a close-fitting fiberboard box conforming to grade W5c of PPP-B-636. All inside faces of the box shall be lined with fiberboard, and each unit container shall be placed closure end uppermost, in a close-fitting cell formed by full-container-height half-slotted fiberboard partitions. Motion of contents within the box shall be prevented by inserting fiberboard pads where needed. Fiberboard lining, partitions and pads shall be formed from material conforming to class weather-resistant grade W5c of PPP-F-320. The box shall be closed in accordance with the applicable requirements of PPP-B-636.

5.1.1.2.2 Five-lb quantity. Six 5-lb unit packs shall be intermediately packed in two rows of three cans per row in the same manner as specified for the 1-lb unit packs in 5.1.1.2.1.

5.1.2 Level B.

5.1.2.1 Unit packing. Potassium hydroxide shall be unit packed level B in a 1-, 5-, or 100-lb quantity as specified (see 6.2).

5.1.2.1.1 One- and five-lb quantity. A 1- or 5-lb quantity as specified (see 6.2) of potassium hydroxide shall be unit packed level B in the same manner as level A, except that if the metal can is used the side-seam need not be striped.

5.1.2.1.2 One hundred-lb quantity. A 100-lb quantity of potassium hydroxide shall be unit packed level B in the same manner as for level A in 5.1.1.1.3, except that the drum shall conform to DOT Specification 21C having a 0.001 inch minimum thickness polyethylene film bonded to the innermost ply and facing the bagged contents.

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5.1.2.2 Intermediate packing. The 1- and 5-lb quantity of potassium hydroxide shall be intermediately packed level B in the same manner as level A in 5.1.1.2 except that the boxes shall conform to class domestic, variety single wall, grade 200 of PPP-B-636.

5.1.3 Level C. A specified quantity (see 6.2) of potassium hydroxide shall be unit and intermediately packed in a manner to assure maintenance of the specified purity and quantity of contents, integrity of containers for a minimum duration of six months, and acceptance by common carrier. Containers shall be in accordance with Uniform Freight Classification Rules, National Motor Freight Classification Rules and the rules imposed by any other intended mode of transportation.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 One-lb quantity. Four intermediate containers of potassium hydroxide of 5.1.1.2.1 shall be packed in two layers of two boxes per layer, with closure ends uppermost, in a close-fitting wood box. The box shall conform to class 3, style optional of PPP-B-585. All wood parts of the box shall be preserved as specified for the grade A box of PPP-B-621. Motion of contents shall be prevented by inserting fiberboard pads where needed. The box shall be closed in accordance with the requirements of PPP-B-585. Fiberboard pads shall be formed from material conforming to class weather-resistant, grade W5c of PPP-F-320.

5.2.1.2 Five-lb quantity. Two intermediate packs of 5-lb unit containers of potassium hydroxide, arranged side by side with closure ends uppermost, shall be packed in the same manner as specified for the intermediate 1-lb unit packs of 5.2.1.1.

5.2.1.3 One hundred-lb quantity. The 100-lb quantity of potassium hydroxide, unit packed as specified in 5.1.1.1.3, shall require no further protection for shipment other than unitization.

5.2.2 Level B.

5.2.2.1 One-lb quantity. The 1-lb quantity intermediate packs shall be packed level B in the same manner as level A, except that the box shall conform to class weather-resistant, grade V3c of PPP-B-636, and the box shall be reinforced as specified therein.

5.2.2.2 Five-lb quantity. The 5-lb quantity intermediate packs of potassium hydroxide shall be packed level B in the same manner as for level A in 5.2.1.2, except that the box shall conform to class weather-resistant, grade V3c of PPP-B-636, and shall be reinforced as specified therein.

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5.2.2.3 One hundred-lb quantity. The 100-lb quantity of potassium hydroxide, unit packed level B as specified in 5.1.2.1.2, shall require no further protection for shipment other than unitization.

5.2.3 Level C. A specified uniform quantity (see 6.2) of potassium hydroxide shall be packed upright in containers in a manner to assure safe delivery to destination and acceptance by common carrier. Containers shall comply with DOT regulations, Uniform Freight Classification Rules, National Motor Freight Classification Rules and the rules applicable to any other intended mode of transportation.

5.2.4 Unitization. For level A and B protection, uniform quantities of the 1- or 5-lb packs shall be palletized in accordance with load type I requirements of MIL-STD-147, while uniform quantities of 100-lb packs shall be palletized in accordance with load type III requirements of MIL-STD-147. Pallets shall conform to type IV, grade B for level A shipments, and type IV, grade A for level B shipments. Wood parts of storage aids for pallet loads of level A containers shall be preserved in the same manner as specified for those of the grade B pallets of NN-P-71. Level C packs shall be palletized to conform to DOT regulations, Uniform Freight Classification Rules, National Motor Freight Classification Rules, and the rules applicable to any other intended mode of transportation, and to assure stability in stacking to a minimum height of 10 feet.

5.3 Marking.

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Marking shall be in accordance with DOT regulations and MIL-STD-129 and shall include the lot or batch number and date of manufacture of the potassium hydroxide and the following precautionary information:

WARNING!

CAUSES SEVERE BURNS TO SKIN AND EYES!

Avoid contact with skin, eyes and clothing.
Do not take internally.
In event of contact with eyes, flush with water for 15 minutes, and call a doctor.
In event of contact with skin, flush with water and wash with vinegar.
When handling, wear goggles or face shield.
Wear safety protective clothing where there is danger of spillages, either of solids or of concentrated solutions.
When making solution, add potassium hydroxide slowly to avoid excess heat and possible spattering.

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6. NOTES

6.1 Intended use. Potassium hydroxide is intended for use in chemical analysis, gas absorption, and electroplating.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Class of potassium hydroxide required (see 1.2),
(c) Level of preservation and packing required (see 5.1 and 5.2),
(d) Quantity to be unit packed level A, B, or C (see 5.1.1.1, 5.1.2.1, and 5.1.3),
(e) Whether one-lb quantity shall be unit packed level A in a wide-mouth glass bottle or metal can (see 5.1.1.1.1), and
(f) Quantity to be packed level C (see 5.2.3).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Sampling and testing precautions. This specification covers inspection of chemical material which is potentially hazardous to personnel. Potassium hydroxide rapidly deliquesces in the air and absorbs carbon dioxide. Contact with air must be avoided as much as possible in order to eliminate errors in analysis. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of this material.

MILITARY INTERESTS:

Custodians:

Army - EA
Navy - SH
Air Force - 68

Review activities:

Army - MD
DLA - GS

Preparing activity:

Army - EA

Project No. 6810-B470

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies and other documents referenced herein.

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